Aayan Akram

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Education

McMaster University — Hamilton, ON, Canada
B.Eng, Computer Engineering — *Expected May 2029*

Experience

Research Assistant - ETF Lab — May 2026 – Present
Faculty of Engineering - McMaster University — *Hamilton, ON, Canada*

- Developed MATLAB algorithms to validate radar-based UAV tracking simulations against real-world flight data
- Performed error analysis and parameter tuning to enhance simulation fidelity
- Leveraged these high-fidelity datasets to develop and train ML models for autonomous drone detection and tracking, improving system reliability

Sustainability Research Intern — August 2024
NGF Europe — *Liverpool, United Kingdom*

- Interviewed 5 executives in key departments to evaluate how automation can reduce energy use, waste, and emissions in glass-cord manufacturing processes, targeting an efficiency increase of 15%
- Analysed batch production workflows, identifying 3 areas where waste-heat recovery improved efficiency
- Investigated 4 alternatives to glass and carbon fibre (e.g., graphene-enhanced fibres, Galvorn), assessing sustainability and performance trade-offs for advanced automotive applications, to improve energy efficiency
- Authored a 7-page findings report and presented 5 strategic recommendations to the executive project team

Accelerator Physics Intern — July 2024 – August 2024
STFC @ Daresbury Labs — *Warrington, United Kingdom*

- Executed routine maintenance on CLARA and VELA particle accelerators, visually inspecting 30 components weekly and supporting technical staff in identifying and resolving 3 potential mechanical issues
- Prepared and verified seal quality on 4 vacuum chambers and joints utilised in RF dipole crab cavities, ensuring strict compliance for installation at the Large Hadron Collider (LHC)
- Simulated accelerator vacuum conditions using Molflow and modeled magnetic fields with Opera2D across 3 materials, developing an applied understanding of beamline design tools to save 10 hours

Systems Engineering Intern — May 2024
Arm — *Manchester, United Kingdom*

- Programmed an ARM Cortex-M4 microcontroller to collect "re-entry" telemetry during 15 drop tests, graphing results in Excel to drive critical design decisions for a satellite prototype
- Developed collision-warning protocols on Arm microcontrollers, defining shared signal strength thresholds and distance criteria of 3m to coordinate 5 satellite prototypes across multiple teams
- Built a secure radio link for a simulated satellite, achieving a 90% message decode success rate over 30m

Simulation Engineering Intern — February 2024
Siemens — *Manchester, United Kingdom*

- Modelled and calibrated digital twins of 3 electromechanical systems using Siemens Simcenter, prioritising thermal and electrical dynamics to precisely align high-fidelity simulations with physical experimental sensor data
- Collaborated on a week-long smart-hotel retrofit project, proposing automation and digital-twin technologies to resolve 4 core customer complaints within a £250,000 budget, improving lifecycle workflows

Projects

For a summary of engineering-related projects I have worked on, check out my online portfolio at:
https://aayanakram.vercel.app

Technical Skills

Programming & OS: C, C++, Python, MATLAB/Simulink, Linux
Embedded: Arduino, ESP32, I2C, MAVLink, PCB Soldering, Pixhawk, Raspberry Pi, SPI, STM32, UART
Software & Tools: Altium, Ardupilot, Autodesk Inventor, LTSpice, OpenCV, Roboflow, Solidworks, YOLOv8